

## Linux Package Management

Software is not a file however you see it a file but it's a package.

Every system has its Package management utility (i.e. in RHEL 8 is dnf – yum)

The vendor of the package is responsible for gathering the content of the package in a single file and packaging it with an extension like .exe for windows systems, .deb for debian based systems (like Ubuntu-kali-parrot) and .rpm for rpm based systems (like centos-fedora).

### Each RPM package is a special archive made up of three components:

- The files installed by the package.
- Information about the package (metadata), such as the name, version, release, and arch; a summary and description of the package; whether it requires other packages to be installed; licensing; a package change log; and other details.
- Scripts that may run when this package is installed, updated, or removed, or are triggered when other packages are installed, updated, or removed.

**RPM package file names consist of four elements (plus the .rpm suffix): name-version-release.architecture:**



in RHEL 6

`coreutils-8.30-4.el8.x86_64.rpm`

Diagram showing the components of the RPM package file name `coreutils-8.30-4.el8.x86_64.rpm`. The name is split into four segments: `coreutils`, `8.30`, `4.el8`, and `x86_64`. Below each segment is a label: 'Name' under `coreutils`, 'Version' under `8.30`, 'Release' under `4.el8`, and 'Arch' under `x86_64`. Vertical lines connect each label to its corresponding segment.

the same in RHEL8

- NAME is one or more words describing the contents (coreutils).
- VERSION is the version number of the original software (8.30).
- RELEASE is the release number of the package based on that version, and is set by the packager, who might not be the original software developer (4.el8).
- ARCH is the processor architecture the package was compiled to run on. noarch indicates that this package's contents are not architecture-specific (as opposed to x86\_64 for 64-bit, aarch64 for 64-bit ARM, and so on).

## **Management Utilities:**

1. Rpm
2. Yum / dnf
3. Subscription

## **Package management with rpm:**

The general form of a query is:

- **rpm -q [select-options] [query-options]**

## **RPM queries: General information about installed packages**

- rpm -qa: List all installed packages
  - rpm -qf FILENAME: Find out what package provides FILENAME
- ```
[user@host ~]$ rpm -qf /etc/yum.repos.d  
redhat-release-8.0-0.39.el8.x86_64
```

## **RPM queries: Information about specific packages**

- rpm -q: List what version of the package is currently installed  
[user@host ~]\$ rpm -q yum  
yum-4.0.9.2-4.el8.noarch
- rpm -qi: Get detailed information about the package
- rpm -ql: List the files installed by the package  
[user@host ~]\$ rpm -ql yum
- rpm -qc: List just the configuration files installed by the package  
[user@host ~]\$ rpm -qc openssh-clients
- rpm -qd: List just the documentation files installed by the package  
[user@host ~]\$ rpm -qd openssh-clients
- rpm -q --scripts: List shell scripts that run before or after the package is installed or removed  
[user@host ~]\$ rpm -q --scripts openssh-server

## **INSTALLING-UNINSTALLING-UPDATE RPM PACKAGES**

```
[root@host ~]# rpm -ivh wonderwidgets-1.0-4.x86_64.rpm >>> install -i for install -v for  
verbose -h for hashes progress bar
```

```
[root@host ~]# rpm -evh wonderwidgets-1.0-4.x86_64.rpm >>> uninstall or remove -e
```

```
[root@host ~]# rpm -Uvh wonderwidgets-1.0-4.x86_64.rpm >>> update -U
```

## **SUMMARY OF RPM QUERY COMMANDS**

Add the -p option to query a package file before installation.

| COMMAND                              | TASK                                                                       |
|--------------------------------------|----------------------------------------------------------------------------|
| <code>rpm -qa</code>                 | List all RPM packages currently installed                                  |
| <code>rpm -q NAME</code>             | Display the version of NAME installed on the system                        |
| <code>rpm -qi NAME</code>            | Display detailed information about a package                               |
| <code>rpm -ql NAME</code>            | List all files included in a package                                       |
| <code>rpm -qc NAME</code>            | List configuration files included in a package                             |
| <code>rpm -qd NAME</code>            | List documentation files included in a package                             |
| <code>rpm -q --changelog NAME</code> | Show a short summary of the reason for a new package release               |
| <code>rpm -q --scripts NAME</code>   | Display the shell scripts run on package installation, upgrade, or removal |

## **MANAGING SOFTWARE PACKAGES WITH YUM**

The low-level rpm command can be used to install packages, but it is not designed to work with package repositories or resolve dependencies from multiple sources automatically.

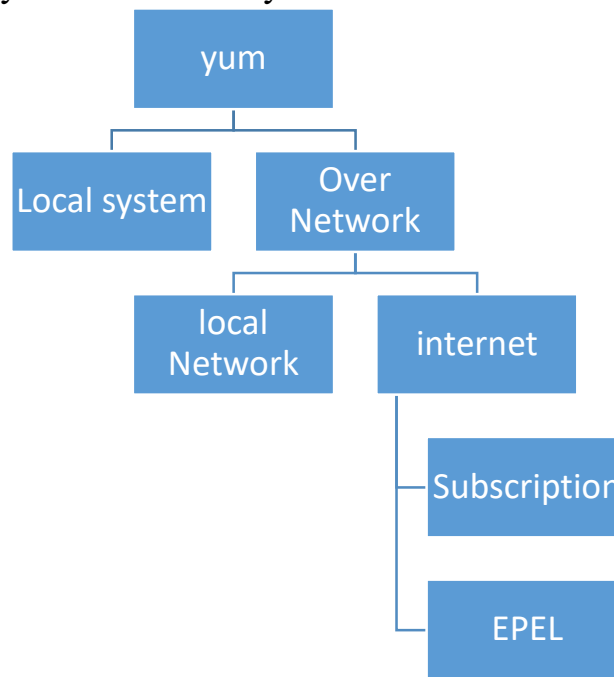
Yum is designed to be a better system for managing RPM-based software installation and updates. The yum command allows you to install, update, remove, and get information about software packages and their dependencies. You can get a history of transactions performed and work with multiple Red Hat and third-party software repositories.

**\*\* yum depend on repo file that you create to locate the repository (server that you store packages in and the metadata or the data about each and every package ) so make sure you create it and it's component right.**

Scenarios to Work with yum :

1. Locally (your system is the yum server and yum client in the same time).

2. Over the Network (your system is client to yum server in the local network or the



internet)

### **Create Local yum repo file:**

1. Mount /dev/sr0 /mnt
2. Vi /etc/yum.repos.d/local.repo >>> and type the following lines

**[Local-BaseOs] >>>>> as the header or ID for repo**

**name = local repo for BaseOS >>>>> as description**

**baseurl = file:///mnt/BaseOS >>>>> as path for packages and it's metadata**

**enabled = 1 >>>>> so you can use this repo for installing and other management operations.**

**gpgcheck = 0 >>>>> don't care about signature**

**[Local-APP] >>>>> as the header or ID for repo**

**name = local repo for AppStream >>>>> as description**

**baseurl = file:///mnt/AppStream >>>>> as path for packages and it's metadata**

**enabled = 1 >>>>> so you can use this repo for installing and other management operations.**

**gpgcheck = 0 >>>>> don't care about signature**

| Option      | Explanation                                                                                                                              |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------|
| [label]     | The label used as an identifier in the repository file.                                                                                  |
| name=       | Specifies the name of the repository.                                                                                                    |
| mirrorlist= | Refers to a URL where information about mirror servers for this server can be obtained. Typically used for big online repositories only. |
| baseurl=    | Refers to the base URL where the RPM packages are found.                                                                                 |
| gpgcheck=   | Set to 1 if a GNU Privacy Guard (GPG) integrity check needs to be performed on the packages. If set to 1, a GPG key is required.         |
| gpgkey=     | Specifies the location of the GPG key that is used to check package integrity.                                                           |

## **Using YUM**

| Task                          | Explanation                                                                        |
|-------------------------------|------------------------------------------------------------------------------------|
| <b>search</b>                 | Search for the exact name of a package                                             |
| <b>[what]provides */name</b>  | Perform a deep search in the package to look for specific files within the package |
| <b>info</b>                   | Provide more information about the package                                         |
| <b>install</b>                | Install the package                                                                |
| <b>remove</b>                 | Remove the package                                                                 |
| <b>list [all   installed]</b> | List all or installed packages                                                     |
| <b>group list</b>             | List package groups                                                                |
| <b>group install</b>          | Install all packages from a group                                                  |
| <b>update</b>                 | Update packages specified                                                          |
| <b>clean all</b>              | Remove all stored metadata                                                         |

## **Examples:**

|                                            |                                   |
|--------------------------------------------|-----------------------------------|
| <b>[root@server3 ~]# yum search user</b>   | <b>find package</b>               |
| <b>[root@server3 ~]# yum info nmap</b>     | <b>getting info about package</b> |
| <b>[root@server3 ~]# yum install nmap</b>  | <b>install package</b>            |
| <b>[root@server3 ~]# yum remove nmap</b>   | <b>uninstall package</b>          |
| <b>[root@server3 ~]# yum list   less</b>   | <b>display list of packages</b>   |
| <b>[root@server1 ~]# yum update kernel</b> | <b>update package</b>             |

## **Working with yum Package Groups**

```
[root@server1 ~]# yum groups list
[root@server1 ~]# yum groups info "Basic Web Server"
```

## **ENABLING RED HAT SOFTWARE REPOSITORIES**

```
[user@host ~]$ yum repolist all
```

```
[user@host ~]$ yum-config-manager --enable rhel-8-server-debug-rpms
```

### **Using EPEL Repository**

```
[user@host ~]$ yum-config-manager --add-repo=http://dl.fedoraproject.org/pub/epel/8/x86\_64/
```

Or

```
[user@host ~]$ yum install http://dl.fedoraproject.org/pub/epel/8/x86\_64/e/epelrelease-8-2.noarch.rpm
```

### **Verify**

```
[user@host ~]$ cat /etc/yum.repos.d/epel.repo
```

To define a repository, but not search it by default, insert the `enabled=0` parameter. Repositories can be enabled and disabled persistently with `yum-config-manager` command or temporarily with `yum` command options, `--enablerepo=PATTERN` and `--disablerepo=PATTERN`.

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